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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	MK GreenCode
Project / Solution Name	Workforce Futures
Name (Leader)	Wong Yik Sheng
Email & Phone (Leader)	yswong2302@gmail.com & +60 11-11682302
University Affiliation(s)	Universiti Malaya
Live Prototype URL	https://workforce-future.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/Griff1018/MK-GreenCode-WorkFuture

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The user works in a browser app (HTML, CSS, vanilla JavaScript) served by a Python FastAPI backend on Replit Deployments. The backend stores accounts, profiles, progress and dated trend snapshots in Replit PostgreSQL through SQLAlchemy.

A refresh job, run daily inside the app or as a Replit scheduled task, calls the Hacker News (Algolia) and arXiv APIs, while job title live from ESCO API, measures how often job posts and research mention each of 39 skill clusters, and saves a snapshot. The scoring engine combines that snapshot with ESCO occupations and skills, the ILO GenAI exposure index and UN SDG targets (bundled as seed data) to rank 20 careers and build reskilling paths.

For reading CVs and rewording explanations, the backend calls the DeepSeek API through its OpenAI-compatible route. On local runs without an API key, the WorkBuddy (Tencent CodeBuddy) CLI can supply the model instead; templates are used when no model is available. Model output is validated before it reaches the user.

Block diagram:

Browser -> Replit: FastAPI app -> PostgreSQL

FastAPI app -> DeepSeek API (local option: WorkBuddy CLI)

FastAPI app -> ESCO API

Daily refresh: Hacker News + arXiv -> dated trend snapshots

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12	Backend language	PSF License
JavaScript, HTML, CSS	Web app front end	Open web standards
FastAPI 0.148.0	HTTP API	MIT
Uvicorn 0.54.0	Web server	BSD-3-Clause
HTTPX 0.28.1	Calls to external APIs and the model	BSD-3-Clause
SQLAlchemy 2.1.5	Database access	MIT
psycopg 3.3.6	PostgreSQL driver	LGPL-3.0
RapidFuzz 3.14.6	Fuzzy matching of job titles and skills	MIT
pypdf 6.21.1	Reading uploaded PDF CVs	BSD-3-Clause
python-docx 1.2.0	Reading uploaded Word CVs	MIT
python-multipart 0.0.32	File uploads	Apache 2.0
openpyxl 3.1.5	Building seed data (development only)	MIT
fpdf2 2.8.9	Generating synthetic sample CVs (development only)	LGPL-3.0
pytest 9.1.1	Automated tests	MIT
GSAP 3.12.5 + ScrollTrigger	Page animations	GSAP Standard License
Lenis 1.1.20	Smooth scrolling	MIT
Newsreader, Figtree (Google Fonts)	Typefaces	SIL Open Font License 1.1
WorkBuddy / CodeBuddy CLI (Tencent)	Optional language-model runner (local runs only)	Tencent CodeBuddy terms of service

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
DeepSeek API (DeepSeek)	Language model: reads CVs, rewords explanations	Paid (pay as you go)
ESCO API (European Commission)	Live job-title search and occupation skills	Free
Hacker News Search API (Algolia)	Job-post demand counts and work headlines	Free
arXiv API (arXiv, Cornell University)	AI research activity per skill	Free
UN SDG API (UN Statistics Division)	Official SDG goal and target text (fetched once, bundled as seed data)	Free
Adzuna API (Adzuna), optional, off by default	Job-ad counts, Singapore as a regional proxy	Free
Google Fonts (Google)	Serves the Newsreader and Figtree typefaces	Free
Unsplash image CDN (Unsplash)	Serves the landing-page background image	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments (Replit)	Hosts the live app. The Repl is linked to the GitHub repository; scripts/build.sh installs dependencies and scripts/start.sh starts Uvicorn	Autoscale deployment / Replit default region
Replit PostgreSQL 16 (Replit)	Accounts, profiles, progress and dated trend snapshots	Included with Replit
Replit Secrets (Replit)	Stores API keys and settings outside the code	Included with Replit
GitHub (GitHub)	Source code and version history	Free
ngrok (ngrok Inc.)	Temporary public link to a locally running copy, used only for testing during development	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
DeepSeek chat model (deepseek-chat)	DeepSeek API	Pre-trained	DeepSeek API terms of service
DeepSeek V4.1 Flash (local runs only)	Tencent CodeBuddy, via the WorkBuddy CLI	Pre-trained	Tencent CodeBuddy terms of service

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ESCO v1.2 occupations and skills	esco.ec.europa.eu	Public	EU reuse policy, attribution	N
ILO Working Paper 140 GenAI exposure scores (Gmyrek et al., 2025)	ilo.org; github.com/pgmyrek/2025_GenAI_scores_ISCO08	Public	CC BY 4.0	N
UN SDG goals and targets	unstats.un.org/sdgs (SDG API)	Public	UN terms of use	N
Hacker News “Who is hiring?” posts (counts only)	hn.algolia.com/api	Public	Public API; only counts stored, never post text or authors	N
arXiv paper metadata (counts only)	arxiv.org (arXiv API)	Public	CC0 (metadata)	N
Adzuna job-ad counts (optional, off by default)	developer.adzuna.com	Public	Adzuna API terms	N
Course and certificate catalogue (71 items)	Each provider’s public page, checked by the team	Self-collected	Each provider’s terms; names and links only	N
TalentCorp MyCOL 2024/2025 (as reported by SAYS, 9 Jan 2026)	says.com	Public	Public report	N
Sample people, teams and CVs	data/seed/samples.json, static/samples/ in the repository	Synthetic	Team-made	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic), Claude Opus 5.5	Writing most of the code from the team's instructions, research checks, design, documentation and submission drafts	Major
Perplexity	Early research draft	Minor
ChatGPT	Prompt refinement	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every AI-generated change was run and checked in the app by the team before it was kept. We added 64 automated tests covering scoring, fairness (age band never changes matches), model guardrails and data checks, and reviewed screenshots of every page in light, dark and phone layouts. Facts, prices and statistics were checked against their original sources, and anything we could not verify was removed or marked as unverified. The team takes full responsibility for the final code and content, and each member can explain the parts they worked on.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Platform (Deployments, PostgreSQL, Secrets)	Hosts the live prototype, its database and its secrets; runs the daily trend refresh

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential to the prototype as built. It hosts the app, its PostgreSQL database and secrets, and runs the daily data refresh. It also provides deploy error fix by free AI Agent. The code itself is standard Python and PostgreSQL, so it could be moved to another cloud host with some setup work. No other sponsor resource like RunPod is used as the language model connected comes from the DeepSeek API, which the team accesses directly.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The app stores, in Replit PostgreSQL, each account's name, profile (skill levels, interests, country, optional age band and situation, weekly hours, budget), chosen career, path progress, resume fields the user types, and chat history with the in-app helper. Session tokens are stored only as SHA-256 hashes. Uploaded CVs are read in memory and not stored. Before any text reaches the language model (the DeepSeek API, or WorkBuddy on local runs), email addresses, phone numbers and IC numbers are removed. Team lists and member profiles stay in the manager's browser; only skill levels are sent to be scored.

Data is kept until the account is reset. The prototype has no password sign-in, so testers are told to use synthetic data only, and the competition demo uses synthetic sample people and CVs. A production version would add proper sign-in and a PDPA-compliant consent and deletion flow.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Thin or skewed trend data: Hacker News hiring threads are global and tech-heavy. Fix: small samples are damped, snapshots are dated, careers show a range across three futures, and Adzuna job ads can be added.
Hallucination: the model only picks skills from a fixed list and must quote the user Fix: any output with a number not in our data is rejected and a template is shown. Careers, courses and scores never come from the model.
Bias: age, gender and ethnicity never enter a score; Fix:an automated test checks that age band cannot change matches.
Provider dependency and rate limits: the app depends on the DeepSeek API for language features;
Fix: if it is unavailable, keyword matching and template explanations keep scores and paths working.
Coverage: ESCO is European, so some Malaysian job titles are missing
Fix: course prices and HRD Corp status must be confirmed with providers.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared



Date: 24 October 2026